

XREAL One Pro Test Protocol — Shadow x YU

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Version: 1.0 · **Prepared:** 2026-07-06 (evening, post-Lora email confirming XREAL One Pro purchase)

Hardware: XREAL One Pro (X1 chip, native 3DoF, X-Prism optics, 57° FOV, 171" virtual display, 120 Hz FHD)

Software backend: Shadow v1.5.12 (github.com/alex-jb/shadow-mentor , MIT, public 2026-07-06)

Rendering pipeline: WebXR + Three.js (browser-native) with Flow Immersive optional presentation layer

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Purpose: define what "the XREAL One Pro test week" actually looks like so Lora and Alex agree on the pass/fail criteria before the hardware lands and before the Yeshiva Dean + Vice-Provost demo date is locked.

Ownership and coordination

- Lora holds the physical device. Amazon delivery ETA: max Saturday 2026-07-11. Device stays at Lora's or Alex's per weekly agreement.
 - Alex owns the software pipeline (Shadow endpoints, Three.js scene graph, Flow Immersive scene JSON).
 - Neither party solo-approves a demo scene for the Dean + Vice-Provost session. Both sign off in writing (email is fine) on each scene's readiness before it appears in the executive demo.
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Phase A — Preflight (2026-07-07 to 2026-07-10, no hardware needed)

While the XREAL One Pro is in transit, Alex builds and validates every render path on desktop Chrome + WebXR emulator. Any failure caught here saves live device time next week.

A1. Flow Immersive scene template

Alex registers `a.flow.gl` (Jason Marsh gave free-tier access post 2026-06-22 call). Builds a scene JSON template that consumes `POST /api/deliberate` output and renders the five personas as spatial cards in a 4x4 meter room, plus the hash-chain audit trail as a walk-around 3D object.

Expected deliverable: `docs/xreal-one-pro-test-protocol/flow-scene-template.json` in this repo before Sat 2026-07-11.

A2. WebXR emulator dry-run

Chrome WebXR API emulator (Chrome DevTools → Sensors → Virtual reality profile) validates: - Scene graph loads without a runtime error - 5 persona cards positioned at $[\pm 1.5\text{m}, 1.6\text{m}, \pm 1.5\text{m}]$ (semicircle

around origin) - Hash-chain object at [0m, 1.4m, -2.0m] (walk-through-able) - FPS overlay ≥ 60 fps on M-series MacBook + Chrome - Camera aligned so a reviewer at [0m, 1.6m, 1m] sees all 5 cards without turning

A3. Fallback rendering paths (defensive plan)

If X1 chip native 3DoF fails at target FPS on-device (documented failure mode in XREAL developer forum posts for large scenes with >800k transparent quads), fall back to:

1. **Reduced-vertex scene** — swap the 3D Gaussian Splatting ambient with a flat cubemap (drops per-frame cost 3× at the cost of losing the immersive background)
2. **Two-persona focus** — hide 3 personas behind a “voice cabinet” UI element; reviewer summons them individually
3. **Tethered mode** — plug One Pro into MacBook Pro USB-C, run scene in Chrome on the laptop, One Pro becomes a 171" virtual display + head-tracking input (X1 chip idle)

Each fallback is coded into the scene graph as a `params.mode = 'full' | 'reduced' | 'focus' | 'tethered'` switch, so Lora and Alex can flip between modes during on-device tests without recompiling.

Phase B — On-device test (starting week of 2026-07-14)

Lora brings the XREAL One Pro to school. Test cycle runs Mon 7/14 → Fri 7/18, one 60-minute session per day.

B1. Session 1 (Mon 2026-07-14) — Standalone rendering baseline

Boot XREAL One Pro standalone (X1 chip, no laptop tether). Load `a.flow.gl` in the device's browser. Open the Phase A1 scene URL. Measure:

- Cold load time (scene JSON fetch + scene graph build)
- Sustained FPS over 3 minutes with reviewer walking around the persona semicircle
- Head-tracking latency (subjective: does the persona card drift with head movement?)
- Battery draw (One Pro internal, per XREAL app)

Pass criteria: ≥ 45 fps sustained, $< 3s$ cold load, no visible head-track drift after 30s.

If any criterion fails: promote reduced-vertex fallback (Phase A3.1), rerun. Log which mode achieved the pass.

B2. Session 2 (Tue 2026-07-15) — Live Shadow endpoint integration

Same scene, but instead of a mocked `/api/deliberate` response, hit the live Shadow endpoint with a real loan scenario. Measure:

- Round-trip latency from persona voice-vote gesture (Web Speech + hand pinch) to verdict-render in the scene

- Attestation verification success rate (client-side `/api/verify-attestation` after each verdict)
- Bill-counsel-facing readability of the `citation` field in each persona's rationale (Lora reads each of the 5 rationales aloud, confirms each cites the correct regulatory anchor)

Pass criteria: < 1.5s round-trip p95, 100% verify success, Lora approves all 5 citation strings for reader-audience clarity.

B3. Session 3 (Wed 2026-07-16) — 4-case verdict lattice

Load each of the 4 canonical case studies from `docs/case-studies/` : - Case 1: \$2.5M CRE loan with PEP owner + diverse routing → escalate - Case 2: First-time HELOC, FICO 640 → block by Credit Fundamentals - Case 3: SBA loan, OFAC SDN hit → block by AML/KYC - Case 4: Standard approve path

For each case, run the scene end-to-end. Measure: - Verdict-render correctness against the case-study expected verdict - 5-voice agreement (do all 5 personas render + speak?) - Gesture-vote submission → hash-chain audit entry (verify via `/api/verify-chain`)

Pass criteria: 4-of-4 verdict correctness, 5-of-5 persona rendering, 4-of-4 audit entries verify.

B4. Session 4 (Thu 2026-07-17) — Executive demo dress rehearsal

Alex + Lora run the full Dean + Vice-Provost demo from start to end, timing the transitions. This is a rehearsal, not a technical test.

- Open scene from `a.flow.gl` URL (30s)
- Play Case 1 (\$2.5M CRE loan) — 90s
- Reviewer walks around the persona semicircle to inspect each verdict (60s)
- Reviewer performs a gesture-vote on the escalation prompt (30s)
- Hash-chain audit trail renders as a walk-through 3D object; reviewer walks through it (60s)
- Q&A anticipation — Lora and Alex rehearse answers to 5 questions the Dean/VP are likely to ask (10 min)

Pass criteria: end-to-end demo runs in under 8 minutes, both Lora and Alex are comfortable with the narrative flow.

B5. Session 5 (Fri 2026-07-18) — Failure mode red-team

Deliberately break the scene in 5 ways and measure recovery: 1. Kill network mid-verdict → does the client retry gracefully, or hang? 2. Send a tampered attestation → does the client-side verifier catch it? 3. Trigger a WebXR permission revoke mid-session → does the scene degrade to 2D safely? 4. Rotate the X1 chip's IMU calibration → does the scene re-anchor? 5. Overload with 10 concurrent verdict requests → does the SIEM's batch-verify (v1.5.12) hold?

Pass criteria: all 5 failure modes recover gracefully. None crash the browser.

Phase C — Dean + Vice-Provost executive demo (target: last week of July or early August, TBC by Lora)

- Lora coordinates the date with the Dean and Vice-Provost office.
 - Alex + Lora demo Case 1 (\$2.5M CRE loan) as the primary narrative. Fallback scenes B2 + B3 are ready if the reviewer asks for a second case.
 - Loom recording (unlisted) of the exact demo goes to the shared Drive within 24h of the session so anyone who missed it can replay.
 - Post-demo debrief: Lora + Alex agree on which scenes to publish (public repo) vs keep private (for procurement-only viewing).
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Failure protocol

If any Phase B session fails a pass criterion: 1. Log the failure in `docs/xreal-one-pro-test-protocol/failure-log.md` 2. Alex has 24h to iterate the scene / endpoint / fallback path 3. Rerun the failed session within 48h 4. Do NOT proceed to Phase C until every Phase B session has a green pass criterion

If a systemic failure (X1 chip driver crash, XREAL SDK regression, etc.): - Alex contacts XREAL developer support - Lora holds the demo date and communicates the slip to the Dean's office - The 4-week Q3 macOS native fallback (per brain memory) becomes the accelerated path

Contact

- Alex Xiaoyu Ji · xji1@mail.yu.edu — software pipeline, scene graph, endpoint integration
 - Loredana C. Levitchi · [email verify] — hardware ownership, executive coordination, banking-domain scene review
 - Jason Marsh · Flow Immersive · [via existing thread] — Flow scene template support if we hit rendering ceilings
 - Hieu Ngo, PhD · Yeshiva Katz School — XR faculty consult, spatial-XR pedagogical scaffolding
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This protocol is a public artifact of the `shadow-mentor` repository (MIT). It is meant to be iterated in-place; any change from either author should be committed with a short “why” note.